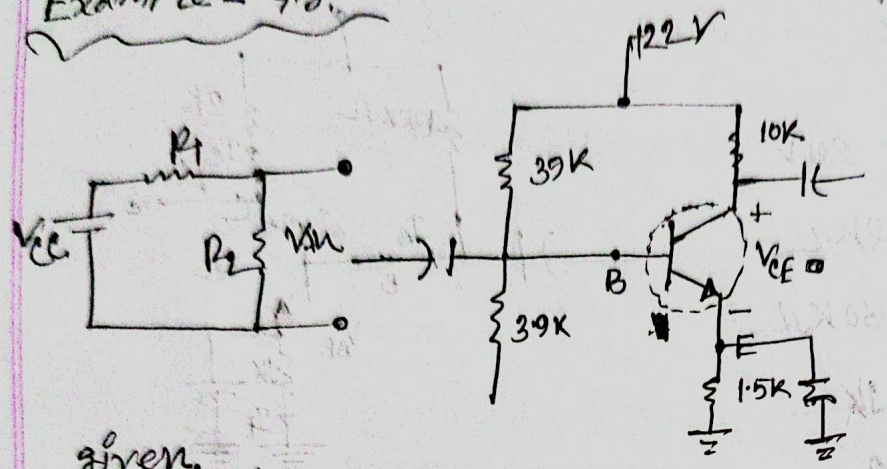
Example – 4.8:



given,

$V_{cc} = 22V$

$R_1 = 39K\Omega$

$R_2 = 3.9K\Omega$

$R_c = 10K\Omega$

$R_E = 1.5K\Omega$

$\beta = 100$

$$R_{th} = R_1 \| R_2 = \frac{R_1R_2}{R_1+R_2}$$

$$= \frac{39 \times 3.9}{3.9+39} = 3.55K\Omega$$

$$E_{th} = V_{th} = \frac{R_2}{R_1+R_2} V_{cc}$$

$$= \frac{3.9}{3.9+39} \times 22$$

$$\therefore E_{th} = 2V$$

$$\therefore I_B = \frac{E_{th} - V_{BE}}{(\beta+1)R_E + R_{th}}$$

$$= \frac{2-0.7}{(100+1)(1.5K\Omega) + 3.55K\Omega} = 8.38\mu A$$

$$\therefore I_c = \beta I_B = 100 \times 8.38 = 0.84mA$$

$$V_{CE} = V_{cc} - R_c(I_c + I_B) = 22 - 10(0.84 + 8.38 \times 10^{-3})$$

$$\therefore V_{CE} = 13.52V.$$